

Practical Tools for Study Design and Inference

Chapter 18: Nonprobability Sampling

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Outline

1. Background

2. Approaches to Inference

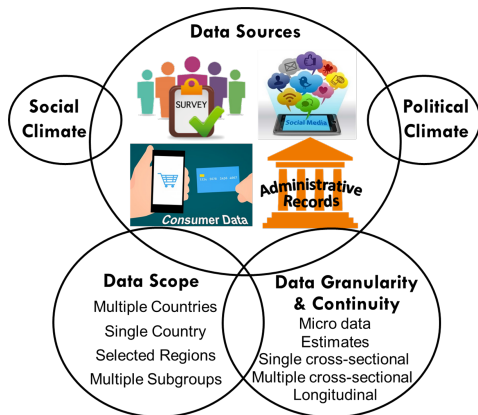
- 2.1 Quasi-Randomization
- 2.2 Super Population
- 2.3 Summary
- 2.4 Exercise

1. Background

History and Examples of Nonprobability Sampling

- Historical – Quota sampling
- Feasibility – Experiments in agricultural research and critical trials
- Historical Examples
 - 1936 Presidential Election poll (Landon vs. Roosevelt) with Literary Digest subscribers
 - 1948 Presidential Election polls (Dewey vs. Truman) by Gallup and Roper using quota samples
 - 2015 British parliamentary election polls and 2016 US Presidential Election polls (Probability sample → Nonprobability)
- Convenient sampling: Using volunteers; e.g., Literary Digest poll, Google Surveys, Opt-in Web panels
- Sample matching: Draw samples that match the population on a set of characteristics; e.g., Quota sampling
- Network sampling: E.g., Snowball sampling, Respondent driven sampling

Emergence of Nontraditional Electronic Data



Pasek & Lee (2022 Fig 1)

- Because of declining response rates and ever increasing costs, pressures to find alternatives to expensive probability sampling have been building.
- Many of these non-traditional databases, despite being large, are not probability samples, but analysts want to project them to full finite populations.

Fit For Purpose: Meeting Goals

- **Relevance**: Statistical information meets user needs?
- **Accuracy**: Statistical information correctly describes the phenomena it was designed to measure?
- **Timeliness**: Delay between the end of the reference period to which statistical information pertains and the date on which the information becomes available.
- **Accessibility**: Ease with which statistical information can be obtained.
- **Coherence**: Statistical information is logically consistent and can be brought together with information from other sources or different time periods?
- **Interpretability**: Availability of supplementary information (metadata) necessary to understand, analyze and utilize statistical information appropriately

Problems with Nonprobability Samples

- Selection bias: Little known about the selection process
- Nonresponse: Not a problem only for probability samples
- Attrition: Esp. Web panels
- Measurement error: Mode effect; Motivated by incentives → satisficing
- Complex data structure
- Uncertain continuity (e.g., MTurk → CloudResearch)

Selection Bias in Nonprobability Sampling - 1

From missing data point of view

For a population parameter vector Θ , its inference based on $\mathbf{Y}_s$ (data observed from a sample s) can be expressed in a likelihood function:

$$L(\Theta; s, \mathbf{Y}_s) \propto \int pr(s|\mathbf{Y}; \Theta) f(\mathbf{y}; \Theta) d\mathbf{Y}_{\bar{s}},$$

where

- $pr(s|\mathbf{Y}; \Theta)$: Selection probability given $\mathbf{Y}$ determined by sample design; Sampling mechanism
- $f(\mathbf{Y}; \Theta)$: Estimation model for $\mathbf{Y}$
- $\mathbf{Y}_{\bar{s}}$: Nonsampled part of $\mathbf{Y}$ in the population
- $\mathbf{Y} = \mathbf{Y}_s + \mathbf{Y}_{\bar{s}} = (Y_1, Y_2, \dots, Y_N)$

Selection Bias in Nonprobability Sampling - 2

- $\mathbf{X}$: $n \times p$ matrix for sample design or covariate in estimation
- $\delta_s = (\delta_1, \delta_2, \dots, \delta_N)$: Vector a 1,0 indicator for a sample s
- Φ : Unknown parameter related to sampling
- Sampling mechanism:

$$pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi)$$

- In probability sampling:

$$\begin{aligned} pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi) &= pr(\delta_s | \mathbf{X}) \text{ or } pr(\delta_s) \text{ in SRS} \\ &\rightarrow L(\Theta; s, \mathbf{Y}_s) \propto \int pr(\delta_s | \mathbf{X}) f(\mathbf{Y} | \mathbf{X}; \Theta) d\mathbf{Y}_{\bar{s}} \end{aligned}$$

- In nonprobability sampling:

$$\begin{aligned} pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi) &\neq pr(\delta_s | \mathbf{X}) \\ &\rightarrow L(\Theta; s, \mathbf{Y}_s) \propto \int pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi) f(\mathbf{Y} | \mathbf{X}; \Theta) d\mathbf{Y}_{\bar{s}} \end{aligned}$$

2. Approaches to Inference

Nonprobability Sampling and Inference

- Design-based inference is "off the table" for nonprobability sampling
- Quasi-randomization approaches
 - Modelling $pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi)$
 - Ideally MAR, leading to $pr(\delta_s | \mathbf{Y}, \mathbf{X}; \Phi) = pr(\delta_s | \mathbf{X}; \Phi)$
 - Psuedo-inclusion probabilities
 - Not targetted to a particular variable of interest
- Superpopulation approaches
 - Modelling $f(\mathbf{Y} | \mathbf{X}; \Theta)$
 - Calibration
 - Multilevel regression and poststratification (MRP)
 - Targetted to a particular variable of interest
- We will denote $\mathbf{X}$ with $\mathbf{X}$ and $\mathbf{Z}$ to denote availability differences in some places but will use $\mathbf{X}$ for brevity

Set-Up: Combine prob+nonprob samples and pop data

	i	Y	X	Z	π
Nonprob Sample	1	✓	✓	✓	?/✓
	2	✓	✓	✓	?/✓
	3	✓	✓	✓	?/✓
Prob Sample	4	?	✓	✓	✓
	5	?	✓	✓	✓
	6	?	✓	✓	✓
Non-Sampled	7	?	?	✓	?
	8	?	?	✓	?
	9	?	?	✓	?
	10	?	?	✓	?
	11	?	?	✓	?
	12	?	?	✓	?
	13	?	?	✓	?
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	N	?	?	✓	?

 Y, X, Z : Random variables π : Selection probability

2.1 Quasi-Randomization

Set-Up: Pseudo-Weights

i	Y	X	Z	π	π_{np}
1	✓	✓	✓	?/✓	$f(\pi_{np} X, Z, \pi_p)$
2	✓	✓	✓	?/✓	$f(\pi_{np} X, Z, \pi_p)$
3	✓	✓	✓	?/✓	$f(\pi_{np} X, Z, \pi_p)$
4	?	✓	✓	✓	
5	?	✓	✓	✓	
6	?	✓	✓	✓	
7	?	?	✓	?	
8	?	?	✓	?	
9	?	?	✓	?	
10	?	?	✓	?	
11	?	?	✓	?	
12	?	?	✓	?	
13	?	?	✓	?	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
N	?	?	✓	?	

- Assumes a hidden selection mechanism for nonprob sample that is estimable with X, Z
- Generate weights that can be used for nonprob sample alone; or if Y available for prob sample, in combination with prob sample.
- No direct modeling of Y given X, Z

Pseudo-Weights - 1

Estimating $pr(\delta_s|\mathbf{X}; \Phi)$ with MAR

- S_i, S_i^* : 1,0 sampling indicator of the prob (S_i) and nonprob sample (S_i^*)

$$\begin{aligned} pr(S_i^* = 1|\mathbf{x}_i) &= \frac{pr(\mathbf{x}_i|S_i^* = 1)pr(S_i^* = 1)}{pr(\mathbf{x}_i)} \\ &\propto \frac{pr(\mathbf{x}_i|S_i^* = 1)}{pr(\mathbf{x}_i|S_i = 1)}pr(S_i = 1|\mathbf{x}_i) \quad \text{through Baye's rule} \end{aligned}$$

- Difficult to estimate $pr(\mathbf{x}_i|S_i = 1)$ and $pr(\mathbf{x}_i|S_i^* = 1)$

→ Use combined prob+nonprob sample with

C_i : 1,0 indicator of nonprob case in the combined data

$$\frac{pr(\mathbf{x}_i|C_i = 1)}{pr(\mathbf{x}_i|C_i = 0)} \propto \frac{pr(C_i = 1|\mathbf{x}_i)}{pr(C_i = 0|\mathbf{x}_i)} = \frac{pr(C_i = 1|\mathbf{x}_i)}{1 - pr(C_i = 1|\mathbf{x}_i)}$$

$$\Rightarrow pr(S_i^* = 1|\mathbf{x}_i) \propto \frac{pr(C_i = 1|\mathbf{x}_i)}{1 - pr(C_i = 1|\mathbf{x}_i)}pr(S_i = 1|\mathbf{x}_i)$$

Pseudo-Weights - 2

Pseudo-Weights, w_i

- When $\mathbf{X}$ contains all variables related to prob sample design, nonresponse and poststratification adjustments, $pr(S_i = 1|\mathbf{x}_i)$ is known

$$\begin{aligned} w_i &= \hat{pr}(S_i^* = 1|\mathbf{x}_i)^{-1} \\ &= \frac{1 - \hat{pr}(C_i = 1|\mathbf{x}_i)}{\hat{pr}(C_i = 1|\mathbf{x}_i)} \frac{1}{pr(S = 1|\mathbf{x}_i)} \end{aligned}$$

where $\hat{pr}(C_i = 1|\mathbf{x}_i)$ estimated from logistic regression

- Otherwise, $pr(S_i = 1|\mathbf{x}_i) = \tilde{w}_i^{-1}$ needs to be estimated (e.g., beta regression of $\mathbf{x}_i$ on w_i)

$$w_i = \frac{1 - \hat{pr}(C_i = 1|\mathbf{x}_i)}{\hat{pr}(C_i = 1|\mathbf{x}_i)} \frac{1}{\hat{pr}(S = 1|\mathbf{x}_i)}$$

Pseudo-Weights - 3

Pseudo-Weights, w_i (cont'd)

- Or estimate $pr(S_i^* = 1|\mathbf{x}_i)$ not conditioned on being in the combined sample
 - Run a weighted logistic regression of the combined sample with a weight of 1 for nonprobability cases and the probability sampling weight for probability sample cases
 $\rightarrow w_i = \hat{pr}(S_i^* = 1|\mathbf{x}_i)^{-1}$
- The models can include $\mathbf{Y}$, if available in prob and nonprob surveys
- For high-dimensional $\mathbf{X}$, consider reduction methods (e.g., LASSO, BART, Superlearner)

Pseudo-Weights - 4

- For point estimates

- If Y available only in nonprob, use w_i (can also be normalized to the population size)
- If Y available in both and if to use the combined sample, normalize weights

- For nonprob cases: $\hat{w}_i = w_i C_{S^*}$, where

$$C_{S^*} = \frac{n_{S^*}}{n_{S^*} + n_S} \frac{\sum I(C_i = 0) \tilde{w}_i}{\sum I(C_i = 1) \tilde{w}_i}$$

(i.e., Weighted fraction of the nonprob sample equals to the unweighted fraction of the nonprob sample in the combined data)

- For prob cases: $\hat{w}_i = \tilde{w}_i C_S$, where

$$C_S = \frac{n_S}{n_{S^*} + n_S}$$

- For variance estimation

- Jackknife estimator: treat nonprob as a single stratum and prob following its own design info

Pseudo-Weights: Example, Lee (2006) - 1

- Nonprob Web sample (s^W):
- Prob ("Reference") sample (s_R):
- Outcome variables
 - y_{blks} : Warm feelings towards blacks
 - y_{vote} : Voting in 2000 presidential election
- Used various unconditional models (e.g., various sets of $\mathbf{X}$) for estimating $pr(S_i^* = 1|\mathbf{x}_i)$
- Created subclasses (c) based on $\hat{pr}(S_i^* = 1|\mathbf{x}_i)$
- Nonprob adjustment weight based on the subclasses:

$$w_i = d_i^W f_c = d_i^W \frac{\sum_{k \in s_c^R} d_k^R / \sum_{k \in s^R} d_k^R}{\sum_{i \in s_c^W} d_i^W / \sum_{i \in s^W} d_i^W}$$

Pseudo-Weights: Example, Lee (2006) - 2

Table 3. Composition of Propensity Score Models^a

Covariate	Propensity score models					
Demographic (D)	D1		D2		D3	
	All covariates		Significant covariates		Nonsignificant covariates	
	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}
Age	✓	✓	✓	✓		
Educ	✓	✓	✓	✓		
.....						
Nondemographic (N)	N1		N2		N3	
	All covariates		Significant covariates		Nonsignificant covariates	
	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}
Class	✓	✓		✓	✓	
Work	✓	✓			✓	✓
.....						
Demographics and nondemographics (A)	A1		A2		A3	
	All covariates		Significant covariates		Nonsignificant covariates	
	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}	<i>y</i> _{blks}	<i>y</i> _{vote}
D1 + N1 D1 + N1 D2 + N2 D2 + N2 D3 + N3 D3 + N3						

^a Included covariates are indicated by check marks.

Note: Propensity Model 4, not shown in the table, is the combination of D1 and N2.

Pseudo-Weights: Example, Lee (2006) - 3

Table 5. Reference Sample, Unadjusted, and Propensity Score Adjusted Web Sample Estimates for y_{blks} and y_{vote}

	$s^{W.ST}$					$s^{W.HI}$				
	estimate	bias	p.bias	se	p.se	estimate	bias	p.bias	se	p.se
y_{blks}										
(M = 2,000)										
y.R	0.612			0.034		0.612			0.034	
y.U	0.636	0.024		0.016		0.675	0.064		0.016	
y.DI	0.623	0.012	52.4%	0.022	38.1%	0.638	0.026	58.6%	0.032	100.0%
y_{vote}										
(M = 1,971)										
y.R	0.650			0.034		0.650			0.034	
y.U	0.715	0.065		0.015		0.817	0.167		0.013	
y.DI	0.709	0.059	9.7%	0.022	46.7%	0.724	0.074	55.7%	0.031	138.5%

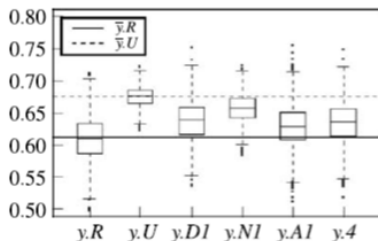
Note: y.R: Reference sample estimate.

y.U: Unadjusted web sample estimate.

y.DI: Web sample estimate after PSA using Model DI.

Pseudo-Weights: Example, Lee (2006) - 4

A: Warm feelings towards blacks



B: Voting participation

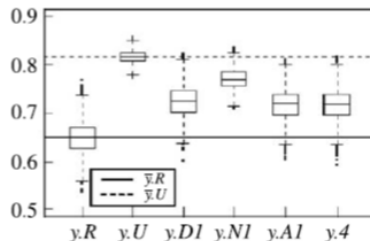


Fig. 4. Distributions of the web estimates by different propensity score adjustments

2.2 Super Population

Set-Up: Calibration

i	Y	X	Z	π
1	✓	✓	✓	?/✓
2	✓	✓	✓	?/✓
3	✓	✓	✓	?/✓
4	?	✓	✓	✓
5	?	✓	✓	✓
6	?	✓	✓	✓
7	?	?	✓	?
8	?	?	✓	?
9	?	?	✓	?
10	?	?	✓	?
11	?	?	✓	?
12	?	?	✓	?
13	?	?	✓	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	?	?	✓	?

- $E_M(y_i | \mathbf{z}_i) = \mathbf{z}_i^T \boldsymbol{\beta}$
- $\hat{y}_i = \mathbf{z}_i^T \hat{\boldsymbol{\beta}}$
- $\hat{t}_y = \sum_{i \in U} \hat{y}_i = \mathbf{t}_{Uz}^T \hat{\boldsymbol{\beta}}$
- $\hat{Y} = \bar{\mathbf{Z}}_U^T \hat{\boldsymbol{\beta}}$, where $\bar{\mathbf{Z}}_U = \frac{\mathbf{t}_{Uz}}{N}$
- Assumes $Y \perp \mathbf{s} | (X, Z)$
- Focus on using nonprob sample.
- If $\mathbf{t}_{Ux}$ is known, x can be incorporated.

Set-Up: Multilevel reg and poststrat (MRP)

i	Y	X	Z	π
1	✓	✓	✓	?/✓
2	✓	✓	✓	?/✓
3	✓	✓	✓	?/✓
4	$f(Y X, Z)$	✓	✓	✓
5	$f(Y X, Z)$	✓	✓	✓
6	$f(Y X, Z)$	✓	✓	✓
7	$f(Y X, Z)$	$f(X Z)$	✓	?
8	$f(Y X, Z)$	$f(X Z)$	✓	?
9	$f(Y X, Z)$	$f(X Z)$	✓	?
10	$f(Y X, Z)$	$f(X Z)$	✓	?
11	$f(Y X, Z)$	$f(X Z)$	✓	?
12	$f(Y X, Z)$	$f(X Z)$	✓	?
13	$f(Y X, Z)$	$f(X Z)$	✓	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	$f(Y X, Z)$	$f(X Z)$	✓	?

- Assumes $(X|Z) \perp s$
- Assumes $f(X|Z)$ is correct
- Assumes $(Y|X, Z) \perp s$
- Assumes $f(Y|X, Z)$ is correct; useful for low-dimension Y
- Allows use of the entire population data; requires rich Z to be plausible

Multilevel Regression & Poststrat. (MRP) - 1

- Poststratification with $h = 1, \dots, H$ poststrata

$$\hat{y}^{PS} = \frac{\sum_h N_h \hat{y}_h}{\sum_h N_h}$$

- Estimate of y at any superpopulation level s (e.g., Michigan)

$$\hat{y}_s^{PS} = \frac{\sum_{h \in H_s} N_h \hat{y}_h}{\sum_{h \in H_s} N_h}$$

- Issues with poststratification: Simple average of poststaratum means
- What if we assume a SRS from a population within each poststratum?
→ Sparse cells; Unstable estimates
- Model-based poststratification using multilevel regression and poststratification (MRP)
 - Bayesian hierarchical model to estimate for sparse poststrata
 - Average over poststrata, weighted by poststrata size in the population

MRP - 2

Consider $Y_i \sim \text{BERNOULLI}(\pi_i)$ (i.e., binary)

K variables forming $h = 1, \dots, H$ poststrata

$$\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \sum_{k=1}^K \beta_{j[i]}^k, \text{ where}$$

$j[i]$ denotes the category of k^{th} variable i unit belongs to

- Priors: $\beta_{j[i]}^k \stackrel{\text{iid}}{\sim} N(0, \sigma_b^2)$; $\sigma_b^2 \sim \text{inv} - \chi^2(\nu, \sigma_0^2)$
- Weak priors for β_0 , ν and σ_0^2
- Use Markov Chain Monte Carlo to get draws from the posterior distribution of β_0 , $\beta_{j[i]}^k$ and σ_b^2
- Use draw of β_0 and $\beta_{j[i]}^k$ to estimate $\hat{\mu}_h = \frac{\sum_{i \in h} \hat{\pi}_i}{n_h}$
- Aggregate $\hat{\mu}_h$ by P_h (proportion of poststratum h in the population)

MRP: Example, Wang et al. (2015) - 1

Wang, W., Rothschild, D., Goel, S., & Gelman, A. (2015). Forecasting elections with non-representative polls. *International Journal of Forecasting*, 31(3), 980-991. (On Canvas)

- Predicting 2012 presidential election outcome from a panel of $n = 350,000$ Xbox users
- $Y_i \sim \text{BERNOULLI}(\pi_i)$: Obama over Romney
- Detailed and highly predictive covariates about voting behaviors: state, education, sex, age, race, party, ideology, 2008 voting behavior
- Prediction model: $\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_{j[i]}^{\text{state}} + \beta_{j[i]}^{\text{edu}} + \beta_{j[i]}^{\text{sex}} + \beta_{j[i]}^{\text{age}} + \beta_{j[i]}^{\text{race}} + \beta_{j[i]}^{\text{party}} + \beta_{j[i]}^{\text{ideology}} + \beta_{j[i]}^{\text{lastvote}} + \text{interactions}$
- Poststratification data: 2008 exit poll with $n=101,638$ (e.g., Proportion of female voters in Michigan who are politically liberal 18-29 year-old Hispanics and voted for Obama in 2008)

MRP: Example, Wang et al. (2015) - 2

Xbox Users vs. 2012 Exit Poll ("truth")

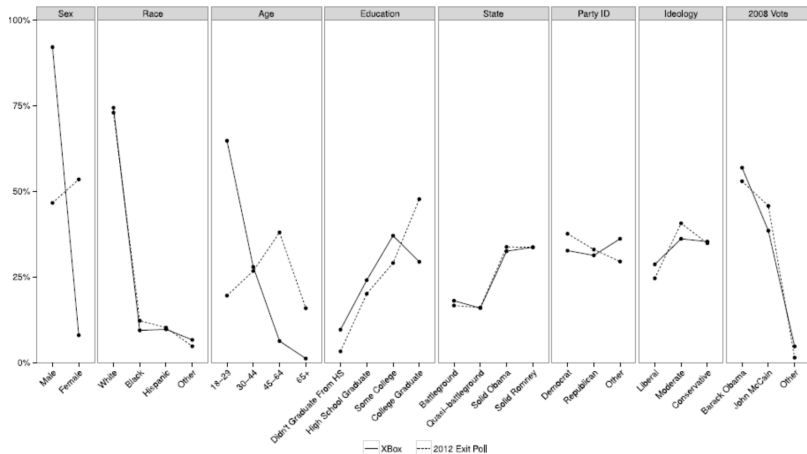


Fig. 1. A comparison of the demographic, partisan, and 2008 vote distributions in the Xbox dataset and the 2012 electorate (as measured by adjusted exit polls). As one might expect, the sex and age distributions exhibit considerable differences.

MRP: Example, Wang et al. (2015) - 3

Xbox Estimates vs. Pollster.com Estimates

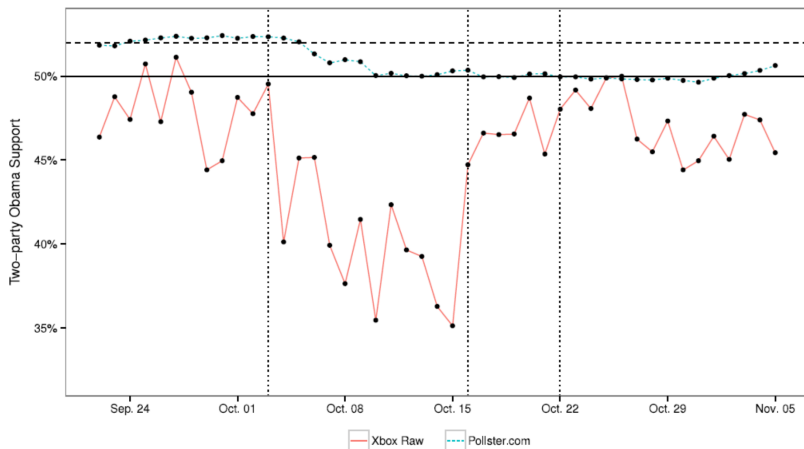
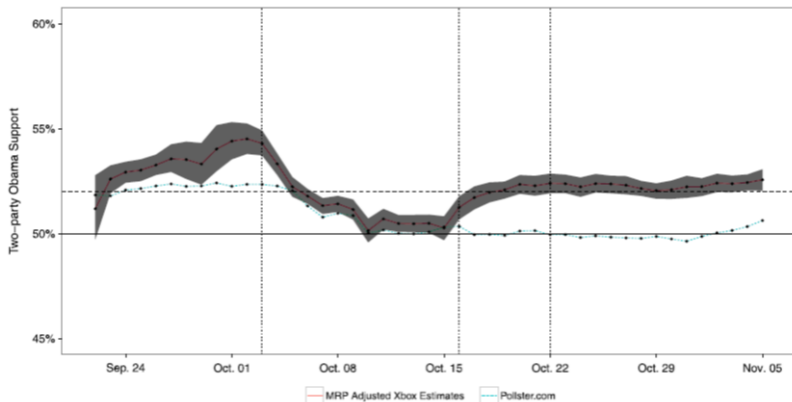


Fig. 2. Daily (unadjusted) Xbox estimates of the two-party Obama support during the 45 days leading up to the 2012 presidential election, which suggest a landslide victory for Mitt Romney. The dotted blue line indicates a consensus average of traditional polls (the daily aggregated polling results from Pollster.com), the horizontal dashed line at 52% indicates the actual two-party vote share obtained by Barack Obama, and the vertical dotted lines give the dates of the three presidential debates. (For the interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

MRP: Example, Wang et al. (2015) - 4

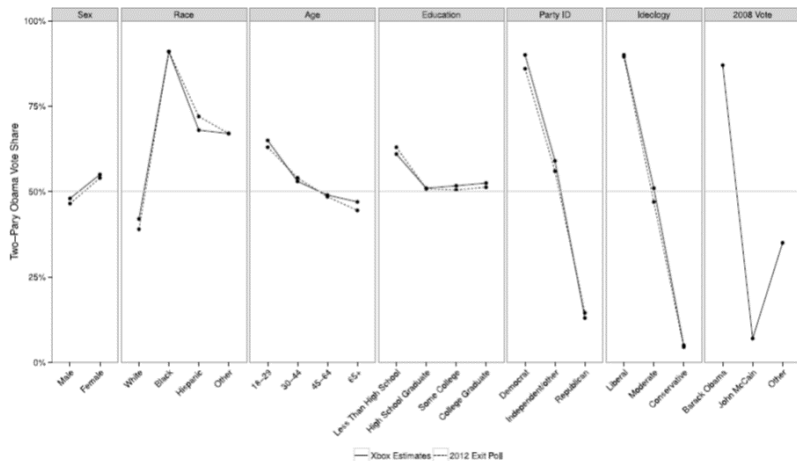
MRP Xbox Estimates vs. Pollster.com Estimates



MRP: Example, Wang et al. (2015) - 5

Secret Sauce!

Xbox vs. 2012 Exit Poll ("truth") on Obama Vote Share



2.3 Summary

Inference of Nonprobability Sample

- Models are necessary
- Quasi-randomization: Convenient; w_i for any y
- Superpopulation: Model specific to each y
- Model specification and richness/relevance of covariates
- Variance estimation requires caution
- Fit for purpose

2.4 Exercise

Pseudo-Weights

- Nonprobability sample data: COVID.csv
- Probability sample data: GSS.csv
- Population data: ACS.csv
- Variable of interest: Proportion of people who rated a statement, "I think politics influenced the development of the COVID-19 vaccines", with moderately agree and strongly agree.

See Exercise Nonprob in Class12Lab.pdf

- What are options for this inference given the data availability?